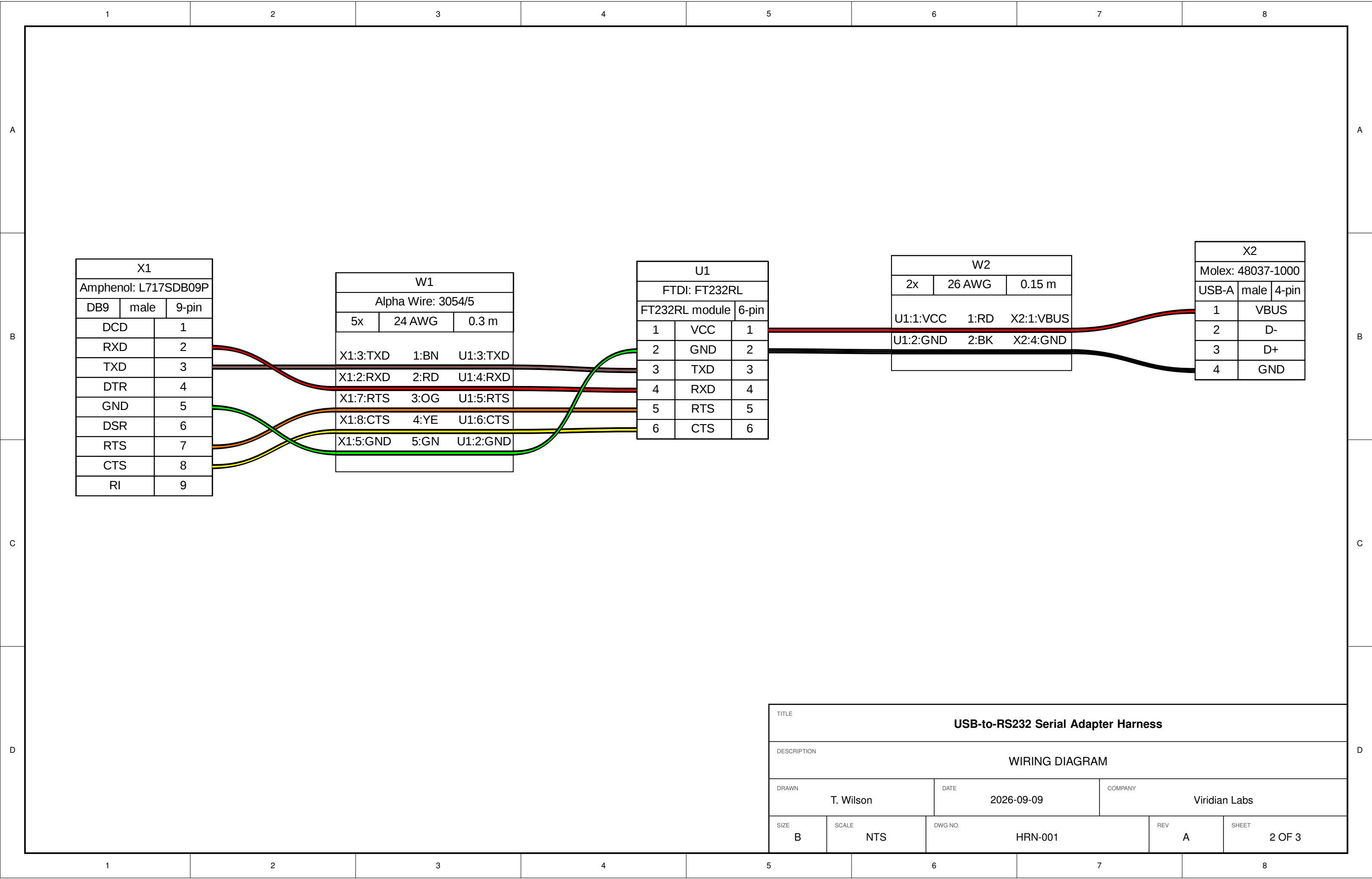


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1 USB-to-RS232 Serial Adapter Harness 1.1 1. Scope Assembly of harness HRN-0042, connecting a DB9 (X1) RS-232 port to a USB-A (X2) host lead through an FT232RL module (U1). Refer to the wiring diagram on sheet 2 and the bill of materials on sheet 3. 1.2 2. Tools & Materials <table><tr><th>Item</th><th>Specification</th></tr><tr><td>Crimp tool</td><td>Molex 63819 series</td></tr><tr><td>Soldering iron</td><td>350 C, chisel tip</td></tr><tr><td>Heat-shrink</td><td>3:1 adhesive-lined, 3 mm</td></tr><tr><td>Flux</td><td>No-clean, rosin</td></tr></table> 1.3 3. Assembly Steps 1. Cut cable W1 to 300 mm and W2 to 150 mm; strip 4 mm on each conductor. 2. Tin every U1 module pad before attaching wires. 3. Solder W1 conductors to U1 per the colour map on sheet 2. 4. Solder W2 RD / BK to U1 VCC / GND. 5. Crimp DB9 (X1) contacts onto W1; seat each contact until the lock tab latches. 6. Terminate USB-A (X2): VBUS to RD, GND to BK. Leave D- / D+ unconnected. 7. Apply heat-shrink over every solder joint and over the U1 module body. 8. Strain-relieve both cable entries with a zip tie 10 mm from the housing. 1.4 4. Test • Continuity-check every pin pair against the bill of materials (sheet 3). • Insulation resistance, X1 shell to all pins: > 20 Mohm at 250 V. • Functional: enumerate on the host, loop TXD-RXD at X1, confirm echo at 115200 baud. Revision A -- initial release.								Item	Specification	Crimp tool	Molex 63819 series	Soldering iron	350 C, chisel tip	Heat-shrink	3:1 adhesive-lined, 3 mm	Flux	No-clean, rosin																																																						
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